

Candidate Name: _____

Class: 07S _____



JURONG JUNIOR COLLEGE

J2 Preliminary Exam 2008

H2 Biology 9747/1

MONDAY

25 AUGUST 2008

1 HOUR 15 MIN

INSTRUCTIONS TO CANDIDATES:

Do NOT open this booklet until you are told to do so.

Write your name and candidate number on the answer sheet in the spaces provided.

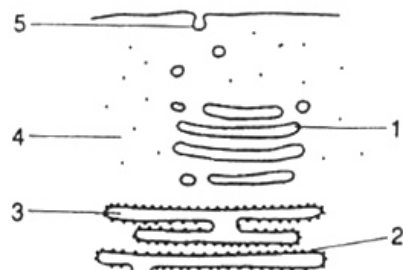
There are forty questions in this paper. Answer all questions. For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

- 1 Radioactive amino acids are supplied to a cell that uses them to make insulin.



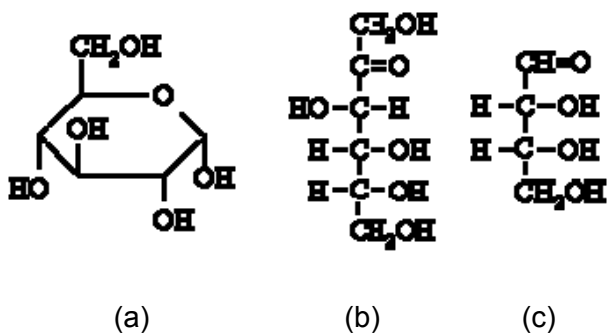
Which route will the radioactive amino acids take?

	First					Last
A	4	2	3	1	5	
B	4	3	2	1	5	
C	5	1	3	2	4	
D	5	3	2	4	1	

- 2 What is the theoretical number of chemically different dipeptides that may be assembled from 12 different amino acids?

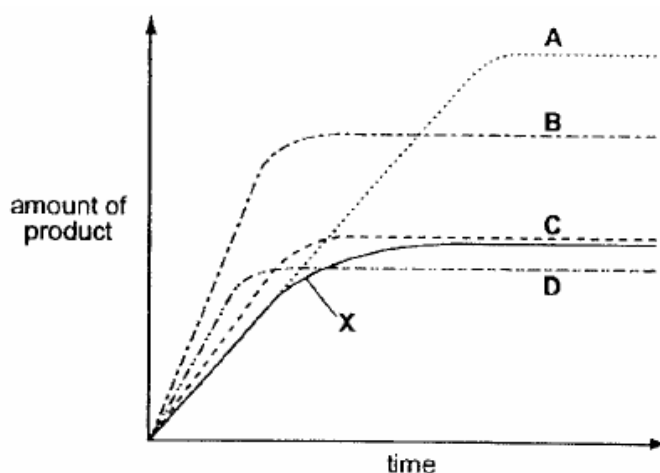
- A 24.
B 72.
C 144.
D 400.

- 3 The structures of some carbohydrates are shown in the diagram below. Which of the following statements is **true**?



- A **a** and **c** are both ketose.
B **b** and **c** are both aldose.
C **a** and **c** are both aldose.
D **b** and **c** are both ketose.

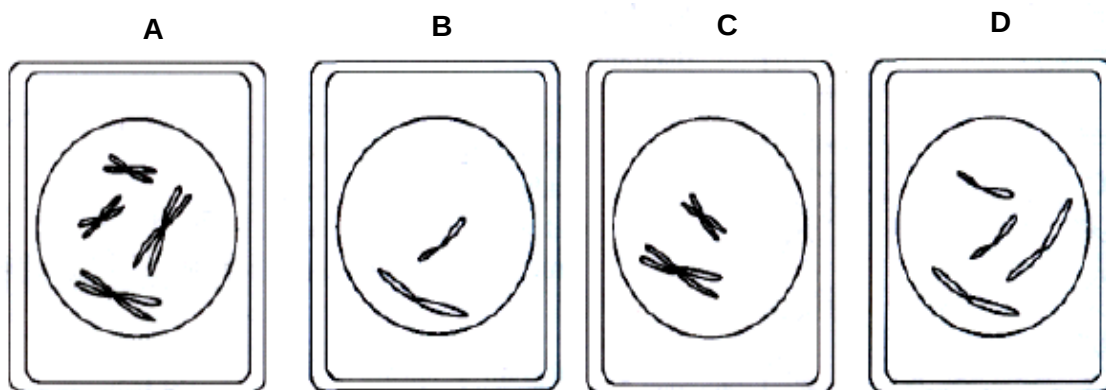
- 4 The curve X shows the activity of an enzyme at 20°C. Curves A to D show the effect of different conditions on the activity of the enzyme. Which curve shows the effect of increasing the temperature by 10°C and adding extra substrate?



- 5 Indinavir[®] is a commercial HIV-1 protease inhibitor administered by doctors to treat AIDS patients. It is supposed to act as a competitive inhibitor. Which of the following shows the correct combination of the K_m value and maximum velocity (V_{max}) of HIV-1 protease when Indinavir[®] is added together with its substrate?

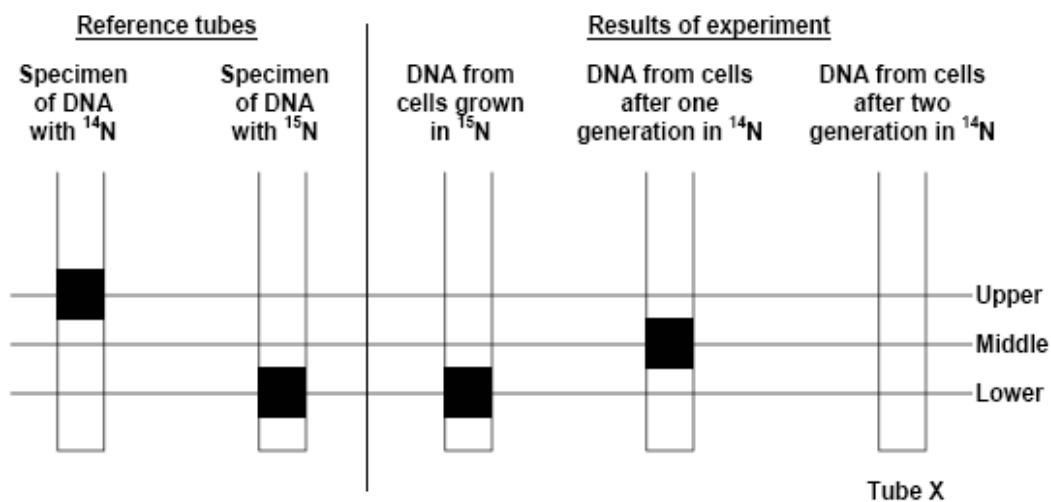
K_m	Maximum velocity (V_{max})
A Remains the same	Increases
B Remains the same	Decreases
C Increases	Remains the same
D Decreases	Decreases

- 6 A diploid cell contains four chromosomes. Which diagram shows the nucleus at prophase of mitosis after a previous meiotic division?



- 7 If DNA is made up of 6 different types of nitrogenous bases coding for 39 types of amino acids in proteins, what would be the minimum number of bases required for each codon?
- A 3
B 5
C 7
D 9
- 8 Cells of the bacterium *E. coli* were grown for many generations on a medium containing only the heavy isotope of nitrogen, ^{15}N . The cells were then transferred to a medium containing only ^{14}N and allowed to grow. Samples of the bacteria were removed from the culture after one generation and after two generations. The DNA from each sample was extracted and centrifuged.

The figure below shows two reference tubes and the results of this experiment.



What are the positions and relative proportions of the bands in Tube X?

	Upper	Middle	Lower
A	50 %	50 %	0 %
B	0 %	50 %	50 %
C	50 %	0 %	50 %
D	25%	50 %	25 %

- 9 You have isolated an antibiotic, edeine, which inhibits protein synthesis but has no effect on either DNA synthesis or RNA synthesis. When added to a translation mixture containing fully intact organelles, edeine stops haemoglobin translation after a short period of time, i.e. 10s.

Analysis of the edeine-inhibited mixture by centrifugation showed that no polyribosomes remained by the time protein synthesis had stopped. Instead, all the mRNA accumulated, together with small ribosomal subunit and initiator tRNA.

Which step in protein synthesis does edeine inhibit?

- A** It prevents the formation of the translation initiation complex, which contains the initiator tRNA and both ribosomal subunits.
- B** It inhibits the binding of aminoacyl-tRNAs to the A-site in the ribosome.
- C** It blocks the translocation of peptidyl-tRNA from the A-site to the P-site of the ribosome.
- D** It interferes with chain termination and release of the peptide.

- 10 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

normal haemoglobin

thr – pro – glu – glu

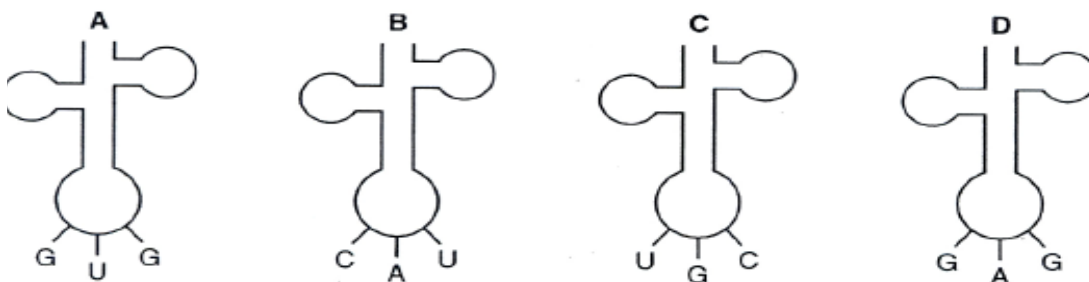
sickle cell haemoglobin

thr – pro – val – glu

mRNA codons for these amino acids are:

name of amino acid	mRNA codons	
glutamine (glu)	GAA	GAG
threonine (thr)	ACU	ACC
proline (pro)	CCU	CCC
valine (val)	GUA	GUG

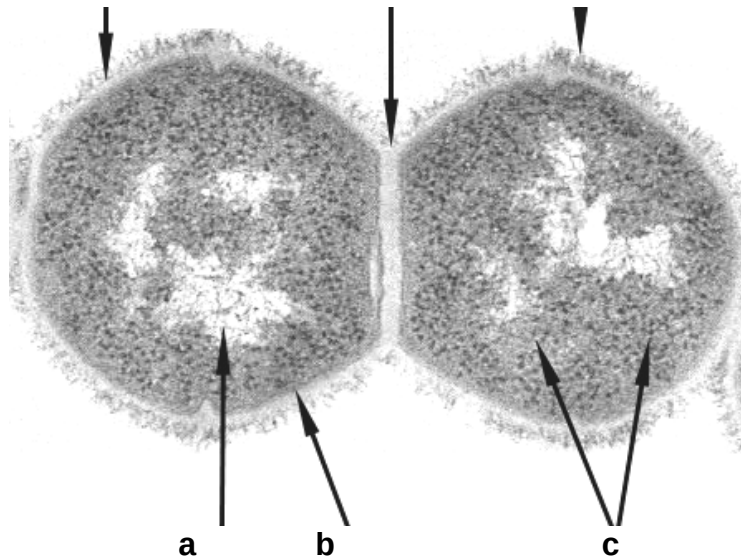
Which transfer RNA molecule is involved in the formation of this part of the sickle cell haemoglobin?



11. Viruses that infect bacteria are called bacteriophages. Each bacteriophage

- A is completely engulfed by a bacterial cell.
- B multiplies outside a bacterial cell.
- C contains DNA and RNA.
- D has a protein coat.

12 The diagram below shows an electron micrograph of a pair of bacteria, *Streptococci*. Which of the following statements is **false**?



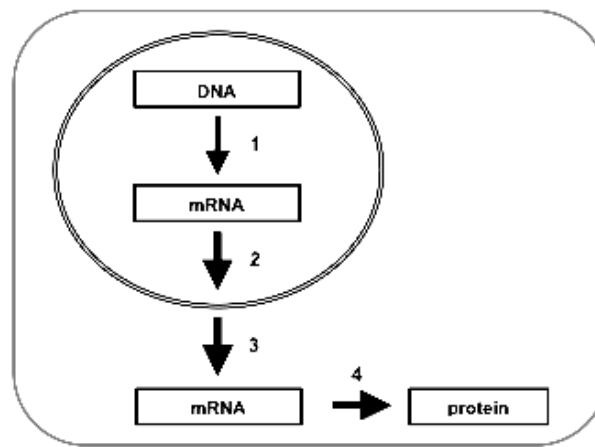
- A Region a contains the genetic material of *Streptococcus*.
- B Region c contains RNA.
- C Both the bacterial cells are genetically identical.
- D The two bacterial cells are undergoing conjugation.

13 If glucose and lactose are both absent from the growth medium, and the *lac* structural genes are expressed efficiently, what would you suspect about the condition of the *E.coli* cell?

- 1 The cell has a mutation in the operator of the *lac* operon.
- 2 The cell has a mutation in the *lacI* gene.
- 3 The cell has a mutation in the promoter of the *lac* structural gene.

- A 1 only
- B 2 only
- C 1 and 2
- D 2 and 3

- 14 The role of telomerase is to maintain the length of telomeres in germline cells. Which statement is **true** about how telomerase works?
- A It has RNA polymerase that can add ribonucleotides to the parental strand.
 - B It has reverse transcriptase that can add deoxyribonucleotides to the parental strand.
 - C It has a DNA template that acts as a template to add deoxyribonucleotides to the parental strand.
 - D It has DNA polymerase that can add deoxyribonucleotides to the parental strand.
- 15 Which of the following describes the events taking place in an eukaryotic cell?



- | | 1 | 2 | 3 | 4 |
|---|-------------------------|----------------|----------------|-----------------------|
| A | gene amplification | mRNA stability | RNA processing | translational control |
| B | gene amplification | RNA processing | mRNA stability | translational control |
| C | transcriptional control | mRNA stability | RNA processing | translational control |
| D | transcriptional control | RNA processing | mRNA stability | translational control |

- 16 In mice, a gene influencing coat colour has three alleles. They are

C : black coat
 c^R : brown coat
 c : white coat

Black coat is dominant to both brown and white coat. A mouse with the genotype $c^R c$ results in another coat colour known as fawn. It is reasonable to predict that the cross

- A $C c^R \times C c^R$ would result in all black mice.
- B $c^R c^R \times c^R c$ would result in all brown mice.
- C $c^R c^R \times c c$ would result in all fawn mice.
- D $C c \times c c$ would result in all white mice.

17 In maize plants, two gene loci on different chromosomes have the following alleles.

Gene locus one

R : green stems

r : red stems

Gene locus two

D : normal height

d : dwarf plants

The following cross was carried out and the seeds collected.

Rr Dd x **Rr Dd**

1600 of the seeds were grown to examine the phenotypes of the offspring.

It would be reasonable to expect that among the offspring, approximately

- A** 100 offspring would be homozygous at both gene loci.
- B** 600 of the offspring will have green stems and be of normal height.
- C** $\frac{2}{3}$ of the green stem, dwarf offspring will be heterozygous at gene locus one
- D** $\frac{1}{3}$ of the red stem, normal height offspring will be heterozygous at both gene loci.

18 In domestic poultry, the all allele for white feathers is dominant to the allele for dark feathers, and the allele for frizzled feathers is dominant to the allele for normal feathers. Poultry which were homozygous for dark, frizzled feathers were bred with poultry bearing white, normal feathers. The offspring of this cross were crossed with homozygous recessive individuals and the following results obtained:

Phenotype	No. of offspring
white, frizzled	18
dark, frizzled	63
white, normal	63
dark, normal	13

The results are due to

- A** gene linkage and crossing over
- B** independent segregation
- C** Mendelian dihybrid inheritance
- D** sex-linkage

- 19 Skin colour in onions is controlled by two pairs of alleles, **Ss** and **Rr**, which segregate independently. The allele **S** is dominant and must be present to allow development of pigment in skin. In its absence, the onion skin is white. Allele **R** is dominant and gives a red colour; the recessive allele **r** gives a yellow colour.

What will be the ratio of phenotypes in the offspring of a cross between plants of genotypes **SsRR** and **ssrr**?

- A** 3 red : 1 yellow
- B** 3 red : 1 white
- C** 1 red : 1 yellow
- D** 1 red : 1 white

- 20 The haemoglobin of sufferers of the hereditary blood disorder sickle cell anaemia has only one amino acid in one of the peptide chains different to normal haemoglobin. The peptide sequence involving the differences are

Normal Haemoglobin: threonine - proline - glutamic acid - glutamic acid - lysine

Sickle cell Haemoglobin: threonine - proline - valine - glutamic acid - lysine

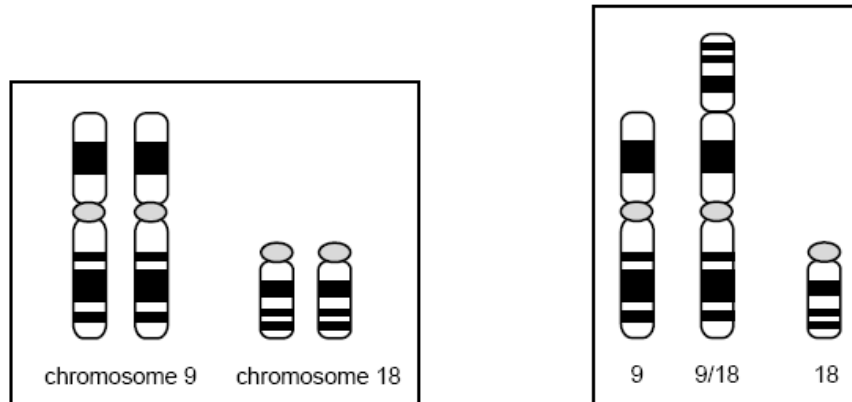
The codons for these amino acids are

Amino acid	Codons
threonine	ACU; ACC; ACA; ACG
proline	CCU; CCC; CCA; CCG
glutamic acid	GAA; GAG
valine	GUU; GUC; GUA; GUG
lysine	AAA; AAG

It is reasonable to conclude that

- A** the sickle cell mutation could have arisen from the eighth base in the DNA sequence of the segment being altered from adenine to uracil.
- B** the anti-codons involved in translation of the sickle cell haemoglobin could be, in order UGA – GGG – CAG – CUC – UUC.
- C** the altered sequence could have arisen during transcription, when the amino acid valine was brought into the growing peptide chain instead of the amino acid glutamic acid.
- D** the DNA sequence of the normal haemoglobin could be, in order TGA – GGG – CAG – CTT – TTT.

- 21 Genetic 'accidents' can occur to chromosomes, and are commonly known as chromosomal mutations. A scientist observed a chromosomal mutation involving chromosomes 9 and 18 in somatic cells of a male cat. The figure below shows chromosomes 9 and 18 in a normal male cat and in a cat with the chromosomal mutation.



The chromosomal mutation involved here is

- A deletion.
- B duplication.
- C inversion.
- D translocation.

- 22 An inhibitor of the enzyme catalyzing the breakdown of citrate is added to respiring tissues. What will be the effect on the concentration of oxaloacetate, pyruvate and ATP?

<i>oxaloacetate</i>	<i>pyruvate</i>	<i>ATP</i>
A decreases	increases	decreases
B decreases	decreases	increases
C increases	increases	decreases
D increases	decreases	increases

- 23 Which statement about respiration is **true**?

- A Cell respiration can only take place in mitochondria.
- B CO₂ is released in the conversion of glucose to pyruvate.
- C Glycolysis takes place in the mitochondrion.
- D Oxygen is the terminal electron acceptor in aerobic respiration.

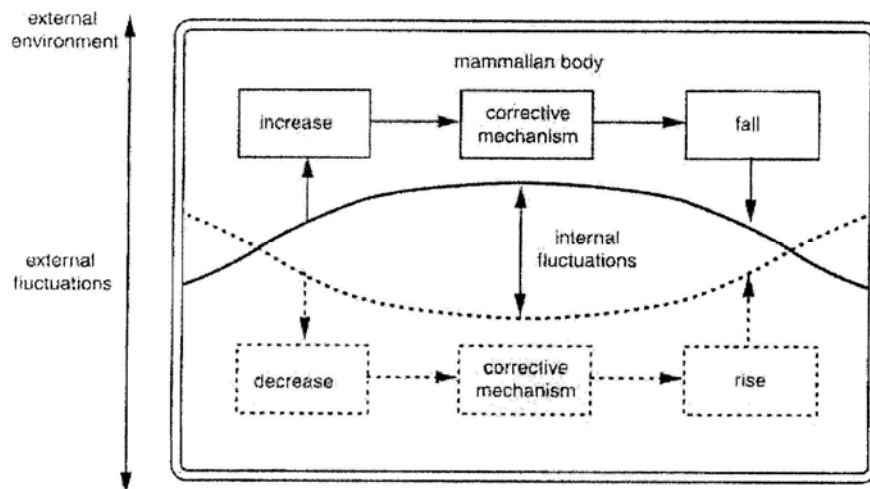
24 Carbohydrates were radioactively labeled and their route from a green plant to a herbivore was identified. Which route was identified?

- | | <i>in the green plant</i> | <i>in the herbivore</i> |
|---|---------------------------|--------------------------------|
| A | glucose → starch | → glucose → maltose → starch |
| B | glucose → starch | → maltose → glucose → glycogen |
| C | starch → maltose | → glucose → maltose → starch |
| D | starch → maltose | → starch → glucose → glycogen |

25 Which process is the link between the light dependent and the light independent stages of photosynthesis?

- A photolysis of water
- B production of carbon dioxide acceptor molecules
- C production of reduced hydrogen carrier and ATP
- D reduction of carbon dioxide

26 The diagram shows some important features of homeostatic mechanisms in the body.



Which of the following correctly describes how a mammal maintains a constant internal environment?

- A When there is a deviation from reference point, the detector detects the deviation and transmits the information to the control center.
- B The control center carries out the response to cancel the deviation, causing fall/rise back to reference point.
- C The effector receives input from the detector and compares the input with the reference point.
- D When reference point is reached, there is feedback information and the mechanism is triggered to enhance the deviation in the opposite direction from reference point.

27 Signal pathways involve components called secondary messengers, characteristics of these substances are as follows:

- i. small
- ii. non-protein
- iii. water-soluble molecules / ions
- iv. enzymatic in nature
- v. membrane bound

Which of the following correctly describes secondary messengers?

- A** i, iii and v
- B** i, ii, iii and v
- C** i, ii and v
- D** i, ii and iii

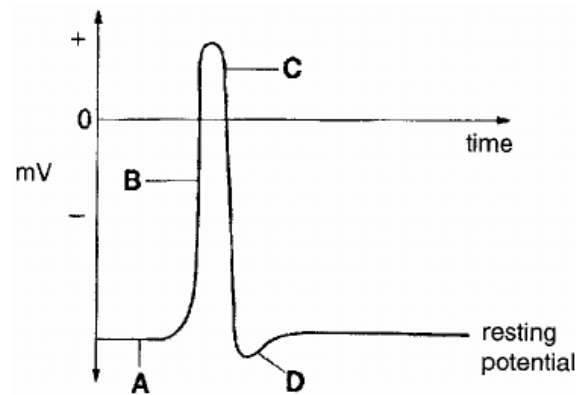
28 When the concentration of glucose in the blood rises above the normal range, the pancreas releases a particular hormone. Some humans cannot make enough of this hormone and must have regular injections of this hormone. Different forms of this hormone have been produced by pharmaceutical companies. The table below shows information on the three forms of the hormone.

Different forms of hormone	Onset of action	Peak action	Duration
P	15 min	30 to 90 min	3 to 5 hours
Q	30 min	2 to 4 hours	6 to 8 hours
R	4 to 8 hours	12 to 18 hours	24 to 28 hours

Which of the following statements is correct?

- A** Form **Q** has a longer duration of action than the other two forms, which means that a person may need fewer injections.
- B** Form **R** has a longer duration than the other two forms, which means that a person may need more injections.
- C** Form **P** is good because a high blood glucose level may be reduced quickly.
- D** Form **Q** is good because a low blood glucose level can be raised quickly as it lasts longer than Form **P**.

29 The diagram shows an action potential. At which point on the graph is the membrane least permeable to sodium ions?



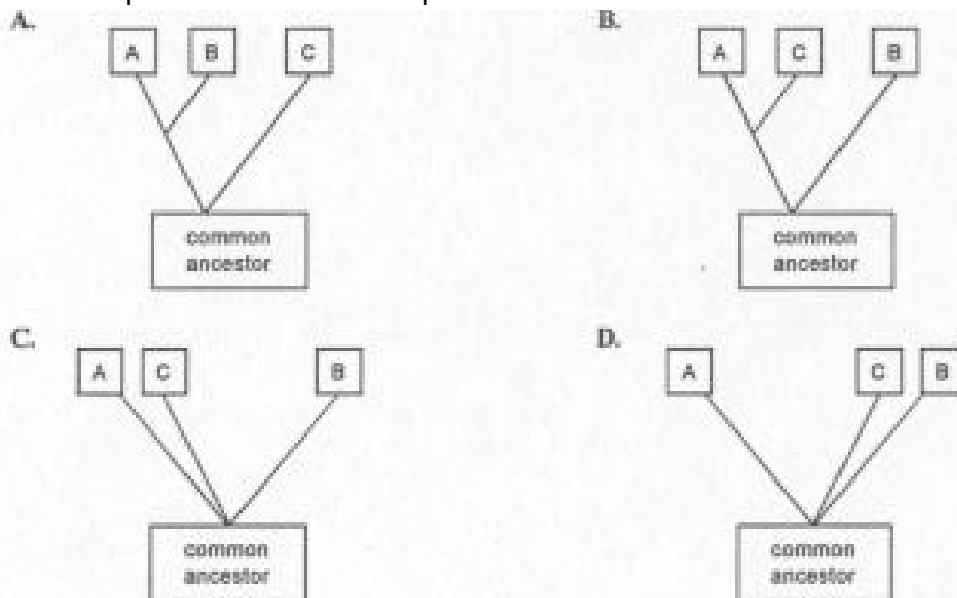
30 For allele frequencies to remain constant in a population, it is essential that

- A mutations occur
- B genetic drift occurs.
- C mating occurs at random in the population.
- D There is regular migration into the population.

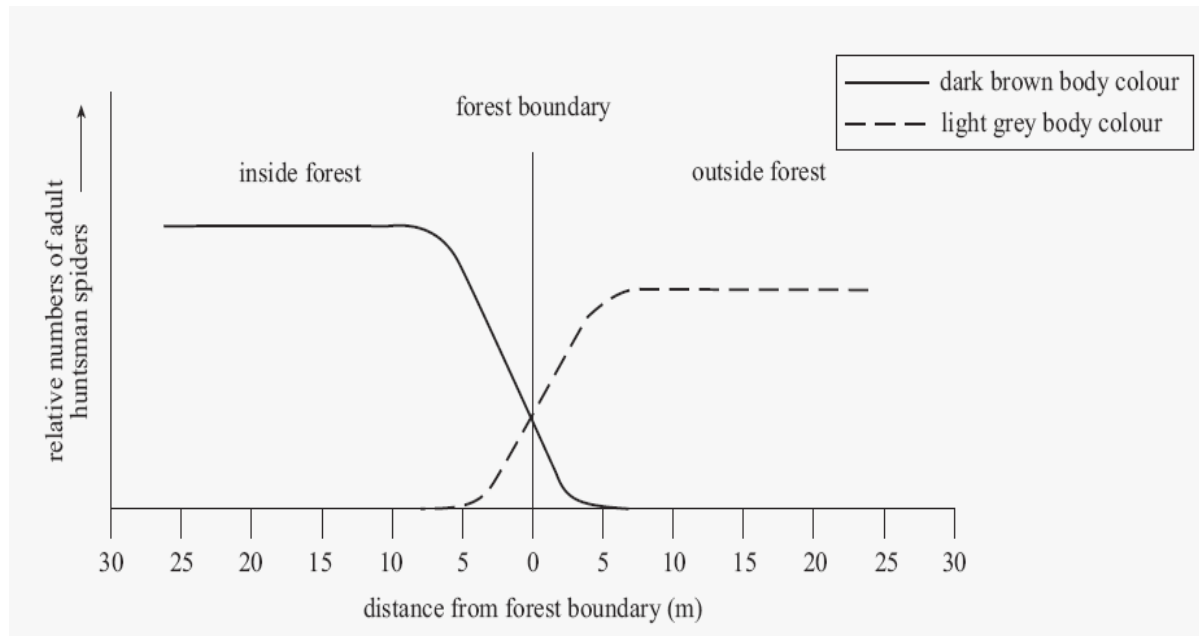
31 Cytochrome c is a protein found in most organisms. The amino acid sequence of this protein varies between species. The numbers of differences in the amino acid sequences in the protein of cytochrome c between three species of chordate A, B and C are shown in the table below.

	Species B	Species C
Species A	11	3
Species B	1234	10

Based on this evidence, the phylogenetic tree that best represents the possible evolutionary relationships between the three species is



32 The graph below shows the distribution of huntsman spiders at a forest boundary:

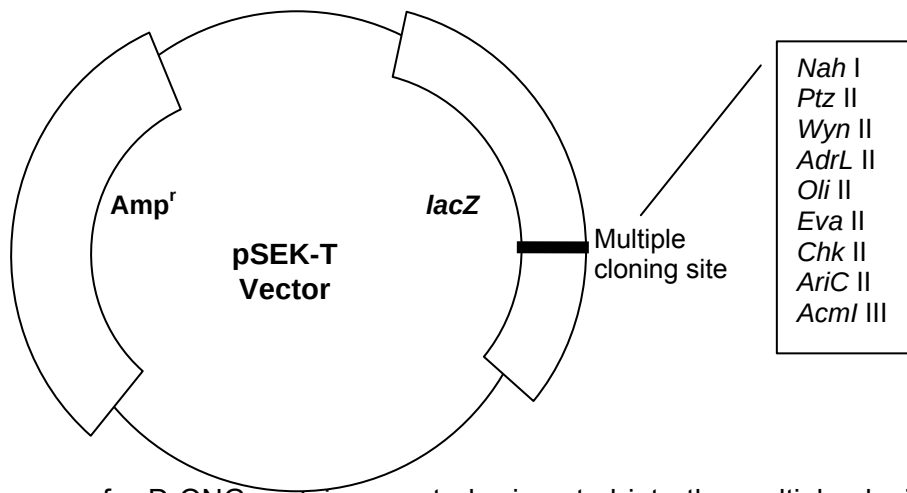


One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

The best explanation for this distribution is that

- A the two populations of spiders were once different species.
- B the two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.
- C the differences in the two habitats had changed the physical appearance of individual spiders.
- D a particular body colour provided a selective advantage to spiders in a particular habitat.

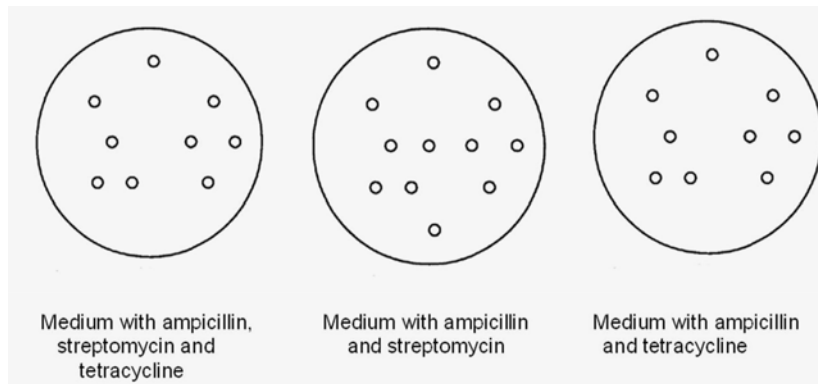
33 As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the D-ONG protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for D-ONG protein were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	Chemicals to be added		Colour of colonies
A	Ampiciliin	X-gal	Blue
B	β -galactosidase	X-gal	Blue
C	Ampiciliin	X-gal	White
D	β -galactosidase	lactose	White

- 34 A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coli* and the bacteria were grown on a nutrient medium with various antibiotics using replica plating. The resulting plates are shown in the diagram.

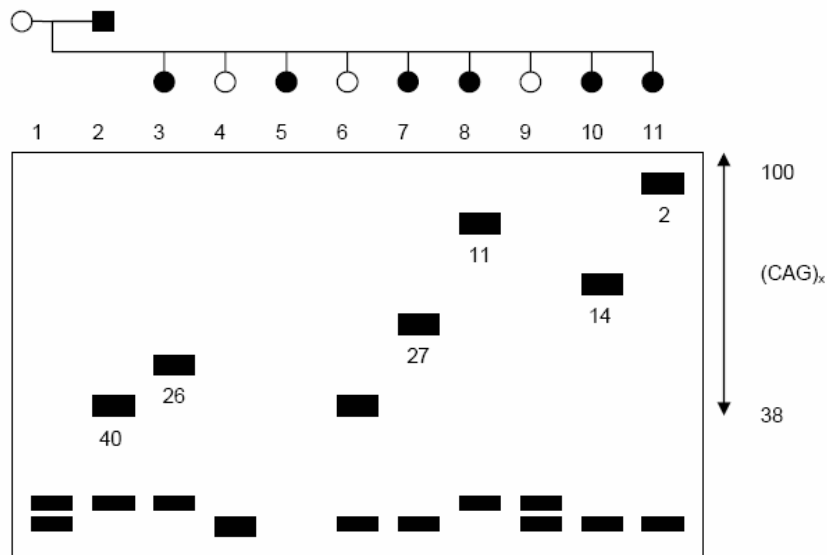


Which antibiotic gene(s) contain(s) the restriction site?

- A Ampicillin
 - B Streptomycin
 - C Tetracycline
 - D Ampicillin and tetracycline
- 35 Plants are more readily manipulated by genetic engineering than are animal cells because
- A plant genes do not contain introns.
 - B more vectors are available for transferring recombinant DNA into plant cells.
 - C a somatic plant cell can often give rise to a complete plant.
 - D plant cells have larger nuclei.
- 36 Which of the following statements are **true** about all stem cells?
- 1 Stem cells can be induced to differentiate by environmental signals.
 - 2 Stem cells are easily isolated and propagated.
 - 3 Stem cells are able to develop into whole organisms if implanted into the womb.
 - 4 Stem cells make more stem cells under appropriate conditions.
- A 1 and 4
 - B 2 and 3
 - C 1, 3 and 4
 - D 1, 2, 3 and 4

- 37 Gene therapy is a biotechnology method that attempts to:
- A sequence a person's genes.
 - B replace a person's mutated genes with normal genes.
 - C produce a DNA fingerprint to identify a person.
 - D make millions of copies of a segment of a person's DNA.
- 38 Which is a factor that does not keep gene therapy from becoming an effective treatment?
- A Short-lived nature of gene therapy.
 - B Problems with viral vector.
 - C Defects at a single locus.
 - D Overexpression of gene.
- 39 Amplification of DNA in the polymerase chain reaction requires
- A nucleotides of uracil.
 - B DNA polymerase.
 - C primase.
 - D ribose sugar.

40 The data below shows the results of electrophoresis of PCR fragments amplified using primers for the site which has been shown to be altered in Huntington's disease. The inherited mutation in the Huntington's disease gene abnormally repeats the nucleotide sequence CAG from 36 to more than 120 times. The male parent, as shown by the black box, got Huntington's disease when he was 40 years old. Six of his children (individuals **3**, **5**, **7**, **8**, **10**, **11**) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.



What is the likely outcome for the normal individuals **4**, **6**, and **9**?

- A** Individuals **4** and **9** do not have the trait, and will not get Huntington's disease, but individual **6** is likely to start the disease when he reaches his father's age of 40.
- B** Individuals **4**, **6**, and **9** have not inherited the defect causing Huntington's disease.
- C** Individuals **4**, **6**, and **9** will still develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.
- D** Two of the three will develop the disease, since it is inherited as a dominant trait, but the data does not allow you to predict which two.

Questions	Answers	Questions	Answers
1	A	21	D
2	C	22	A
3	C	23	D
4	B	24	B
5	C	25	C
6	C	26	A
7	A	27	D
8	A	28	C
9	A	29	D
10	B	30	C
11	D	31	B
12	D	32	D
13	C	33	C
14	B	34	C
15	D	35	C
16	C	36	A
17	C	37	B
18	A	38	C
19	D	39	B
20	B	40	A